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Publication KR 2024-0168593 A · 2024-12-02 Title (EN) Object attribute based watermarking method for preventing leakage of digital content Applicant Korea Electronics Technology Institute (KETI) Inventors Hong Ji Soo; Hong Sung Hee; Kim Young Min; Jeong Jin Soo; Kim Yong Hwa; Lee Byoung Hyo; Oh Hyeon Chan Priority / application KR 20230066130 A · 2023-05-23 IPC G06F 21/16; G06T 1/00; H04N 21/2389; H04N 21/8358 Abstract (English, as displayed by Espacenet) A watermarking method based on object properties for preventing digital content leakage is provided. A video watermarking method according to an embodiment of the present invention includes acquiring a requested video and watermarking information when a video streaming request is received from a user, determining some objects appearing in the video based on the watermarking information, and modifies the properties of the determined objects and streams them. This provides video content by watermarking the properties of objects appearing in the video differently for each user, thereby making watermark recognition difficult while being robust against various types of attacks and enabling the identification of content leakage routes. Description (English machine translation) Global Dossier EN KO Data originating from sources other than the EPO may not be accurate, complete, or up to date. The wording below is an initial machine translation of the original publication. To generate a version using the latest translation technology, go to the original language text and use Patent Translate. [0001] [The present invention relates to a technology for protecting digital content copyrights, and more specifically, to a watermarking method capable of preventing digital content leakage and tracing the leakage route in the event of leakage.] [0003] [Watermarking technology to prevent leakage of existing digital cinema, video, etc. content has been developed in various ways to protect content copyright, and in particular, a technology is being developed that can protect copyright in a way that provides services to users while allowing the identification of content leakage routes in the future through imperceptible watermarking technology that makes it impossible to recognize the watermark.] [0004] [These watermarking technologies must be robust, especially to be able to authenticate and maintain integrity against various attacks. However, existing technologies use various characteristics of the image to embed user codes, and although there are techniques for achieving robustness, there is a disadvantage that they can be vulnerable to new types of attacks in terms of time and space.] [0006] [The present invention has been made to solve the above problems, and the purpose of the present invention is to provide a non-perceptual watermarking method capable of identifying a video leakage path based on the properties of objects appearing in the video.] [0008] [A video watermarking method according to one embodiment of the present invention for achieving the above object includes the steps of: receiving a request for video streaming from a user; acquiring the requested video and watermarking information; determining some of the objects appearing in the video according to the watermarking information; modifying properties of the determined objects; and streaming the modified video.] [0009] [Objects can be background objects.] [0010] [The decision step may be to determine some background objects based on user code.] [0011] [The decision step may be to determine the background object to be transformed from the user code based on a code matching rule that matches the background object to be transformed according to the user code.] [0012] [The transformation step may be transforming at least one of color and texture.] [0013] [The type of property to be transformed for the background object may be determined based on the user code.] [0014] [The video watermarking method according to the present invention further includes a step of encoding a user ID to generate a user code; and the determining step can use the user code generated in the generating step.] [0015] [A video watermarking method according to the present invention may further include a step of acquiring watermarking information of a leaked video; a step of inferring a user code applied to attribute modification of a background object appearing in the leaked video based on the watermarking information; and a step of decoding the inferred user code to acquire a user ID.] [0016] [The user code inference step may include: a step of determining some background objects whose properties have been modified in the leaked video by referring to watermarking information; a step of inferring a user code from the determination result by referring to a code matching rule.] [0017] [According to another aspect of the present invention, a video streaming server is provided, characterized by including: a communication unit that receives a video streaming request from a user and streams the requested video; a DB in which video and watermarking information are stored; and a processor that acquires the video and watermarking information requested from the user from the DB, determines some objects among objects appearing in the video according to the watermarking information, and modifies properties of the determined objects and streams them through the communication unit.] [0018] [According to another aspect of the present invention, a video leak path tracking method is provided, including the steps of: acquiring watermarking information of a leaked video; identifying a deformed background object in the leaked video by referring to the watermarking information; inferring a user code from the identification result; and decoding the inferred user code to identify a user ID.] [0019] [According to another aspect of the present invention, a video leak path tracking system is provided, including a DB storing a video and watermarking information; and a processor that acquires a leaked video and watermarking information from the DB, determines a deformed background object in the leaked video by referring to the acquired watermarking information, infers a user code from the determination result, and decodes the inferred user code to obtain a user ID.] [0021] [As described above, according to embodiments of the present invention, by providing video content by watermarking the properties of objects appearing in the video differently for each user, it is possible to make watermark recognition difficult while being robust against various types of attacks and to identify the leak path of the content.] [0023] [Figures 1 and 2 are drawings provided for explaining the concept of a video watermarking method according to one embodiment of the present invention, Figure 3 is a drawing showing a video watermarking method in a video streaming server according to another embodiment of the present invention, Figure 4 is a drawing provided for explaining a video leak path tracking method in a video streaming server according to another embodiment of the present invention, and Figure 5 is a drawing showing a hardware configuration of a video streaming server according to another embodiment of the present invention.] [0024] [Hereinafter, the present invention will be described in more detail with reference to the drawings.] [0025] [We present an object property-based watermarking method for preventing digital video content leakage. It is a technology that watermarks the properties of background objects appearing in a video differently for each user and streams them so that watermark recognition is difficult, it is robust against various types of attacks, and the leakage path of the content can be identified.] [0026] [Figures 1 and 2 are drawings provided for conceptual explanation of a video watermarking method according to one embodiment of the present invention. In the embodiment of the present invention, watermarking is performed by modifying attributes of a background object, which is an unimportant portion of video content, differently for each user (viewer), rather than a foreground object, which is an important portion that viewers pay attention to.] [0027] [Specifically, even if users A and B are watching the same scene of the video, the scene that user A is watching (shown in Fig. 1) and the scene that user B is watching (shown in Fig. 2) are different. The only difference is the properties (color and texture in Figs. 1 and 2) of background objects (cups and cushions in Figs. 1 and 2) that do not affect the user's enjoyment of the video and comprehension of the story at all.] [0028] [Since the part of the video content that is transformed is a background object, users can watch the video without being aware of the watermark applied to the video. In addition, since only properties such as color and texture are transformed without changing the position or shape of the background object, the video content itself is hardly damaged, and it can be robust against various attacks.] [0029] [The watermark applied differently for each user, that is, the properties of the background object transformed differently for each user, are used to track the leak path, that is, the user who leaked the video, in the event of a future video content leak.] [0030] [Figure 3 is a diagram illustrating a video watermarking method in a video streaming server according to another embodiment of the present invention.] [0031] [As shown, when a user requests video content to be viewed from a streaming server (100), the streaming server (100) obtains the video requested by the user and the information required for watermarking the video from the DB (110).] [0032] [The information required for watermarking (hereinafter referred to as ‘watermarking information’) consists of a code matching rule and a list of background objects, and is defined differently for each video.] [0033] [The background object list contains information (Object info. #1, Object info. #2, ...) about background objects that can be modified for watermarking in video content. As mentioned above, the background objects are those that can be modified without any problems in viewing the video or understanding the story. The information about the background objects includes the following information.] [0034] [1) [Time information when the background object appears] [0035] [2) [Location information of the background object] [0036] [3) The type (class) of the background object] [0037] [4) [Attribute information to be converted for the corresponding background object] [0038] [The properties to be converted for the background object may include the color, texture, etc. of the background object. In order to be robust against various attacks, the property values to be converted need to be defined so that they differ greatly from the original property values. For example, if the color of the background object is red, the color to be converted is not defined as a similar color such as pink, but as a different color such as blue or green.] [0039] [The ID encoder (120) encodes the user ID to generate a user-specific code. The code matching rule is a rule for determining the background object to be transformed based on the user code.] [0040] [For example, if the user code is assumed to be a 4-digit number (abcd), the code matching rule determines the 'ab'th background object information and the 'cd'th background object. Specifically, if the user code is "0513", information about the 05th background object (Object info. #05) and information about the 13th background object (Object info. #13) are determined.] [0041] [The video transformation unit (130) watermarks the background objects in the video by transforming them with reference to the information about the determined background objects. Accordingly, a video with some of the background objects transformed is streamed to the user. For natural transformation, the video transformation unit (130) can utilize a deep learning algorithm or an image processing technique.] [0042] [Figure 4 is a drawing provided to explain a method for tracing a video leak path in a video streaming server (100) according to another embodiment of the present invention.] [0043] [In order to trace the leak path when a streaming video is illegally leaked, first, the watermarking information of the leaked video is acquired from DB (110). This obtains the code matching rule and background object list of the leaked video.] [0044] [After that, the attribute checker (140) identifies the attributes of the background objects listed in the background object list to check if there is a deformation and determines the background objects whose attributes have been deformed.] [0045] [The following streaming server (100) infers the user code from the determination result of the attribute verifier (140) by referring to the code matching rule. For example, if the background objects whose attributes have been modified are the 05th background object and the 13th background object, and the code matching rule is a rule that determines the background object information corresponding to the first two digits and the last two digits of the user code, the user code can be inferred as "0513".] [0046] [Afterwards, the ID decoder (150) decodes the inferred user code to restore the user ID. The restored user ID can be treated as the ID of the user who leaked the video.] [0047] [Figure 5 is a diagram showing the hardware configuration of a video streaming server (100) according to another embodiment of the present invention. The video streaming server (100) according to the embodiment of the present invention is configured to include a communication unit (101), a processor (102), and a DB (103) as illustrated.] [0048] [The communication unit (101) is a communication means that receives a user's request and streams the video content requested by the user.] [0049] [The processor (102) performs procedures such as user ID encoding/decoding, video transformation (watermarking), and attribute verification required for the watermarking process presented in FIG. 3 and the leak path tracking process presented in FIG. 4.] [0050] [DB(103) is a storage where video and watermarking information are stored.] [0051] [So far, we have described in detail preferred embodiments of object property-based watermarking methods for preventing digital content leakage.] [0052] [In an embodiment of the present invention, a new watermarking method is proposed that provides video content with different object properties depending on the user, thereby making watermark recognition difficult, being robust against various types of attacks, and enabling the identification of content leakage paths.] [0053] [In the above embodiment, the background objects to be transformed are determined through the code matching rule from the user code, but additional transformations are possible. For example, the type of property to be transformed for the background object can also be determined through the code matching rule from the user code. Accordingly, only the color of the background object can be changed, or only the texture can be changed, depending on the user code, and further, both the color and the texture can be changed.] [0054] [Meanwhile, it goes without saying that the technical idea of the present invention can be applied to a computer-readable recording medium containing a computer program that performs the functions of the device and method according to the present embodiment. In addition, the technical idea according to various embodiments of the present invention can be implemented in the form of a computer-readable code recorded on a computer-readable recording medium. The computer-readable recording medium can be any data storage device that can be read by a computer and store data. For example, the computer-readable recording medium can be a ROM, a RAM, a CD-ROM, a magnetic tape, a floppy disk, an optical disk, a hard disk drive, etc. In addition, the computer-readable code or program stored on the computer-readable recording medium can be transmitted through a network connected between computers.] [0055] [Also, although the preferred embodiments of the present invention have been illustrated and described above, the present invention is not limited to the specific embodiments described above, and various modifications may be made by a person skilled in the art without departing from the gist of the present invention as claimed in the claims, and such modifications should not be individually understood from the technical idea or prospect of the present invention.] [0057] [100 : Streaming server 110 : DB 120 : ID encoder 130 : Video transformer 140 : Attribute checker 150 : ID decoder] Claims (English machine translation) Global Dossier EN KO Data originating from sources other than the EPO may not be accurate, complete, or up to date. The wording below is an initial machine translation of the original publication. To generate a version using the latest translation technology, go to the original language text and use Patent Translate. 1. A video watermarking method, comprising: a step of receiving a video streaming request from a user; a step of acquiring the requested video and watermarking information; a step of determining some of objects appearing in the video according to the watermarking information; a step of modifying properties of the determined objects; and a step of streaming the modified video. 2. A video watermarking method according to claim 1, characterized in that the objects are background objects. 3. A video watermarking method according to claim 2, characterized in that the determining step determines some background objects based on a user code. 4. A video watermarking method according to claim 3, characterized in that the determining step determines the background object to be transformed from the user code based on a code matching rule in which the background object to be transformed is matched according to the user code. 5. A video watermarking method according to claim 4, characterized in that the transformation step transforms at least one of color and texture. 6. A video watermarking method according to claim 5, characterized in that the type of property to be transformed for a background object is determined based on a user code. 7. A video watermarking method according to claim 3, further comprising a step of encoding a user ID to generate a user code; and wherein the determining step uses the user code generated in the generating step. 8. A video watermarking method according to claim 7, further comprising: a step of acquiring watermarking information of a leaked video; a step of inferring a user code applied to attribute modification of a background object appearing in the leaked video based on the watermarking information; and a step of decoding the inferred user code to acquire a user ID. 9. A video watermarking method according to claim 8, characterized in that the user code inference step includes: a step of determining some background objects whose properties have been modified in the leaked video by referring to watermarking information; and a step of inferring a user code from the determination result by referring to a code matching rule. 10. A video streaming server comprising: a communication unit that receives a video streaming request from a user and streams the requested video; a DB in which video and watermarking information are stored;

and a processor that acquires the video and watermarking information requested from the user from the DB, determines some objects among objects appearing in the video according to the watermarking information, and modifies properties of the determined objects and streams them through the communication unit. 11. A method for tracking a video leak path, comprising: a step of acquiring watermarking information of a leaked video; a step of determining a deformed background object in the leaked video by referring to the watermarking information; a step of inferring a user code from the determination result; and a step of decoding the inferred user code to obtain a user ID. 12. A video leak path tracking system, comprising: a DB storing video and watermarking information; and a processor obtaining leaked video and watermarking information from the DB, determining a deformed background object in the leaked video by referring to the obtained watermarking information, inferring a user code from the determination result, and decoding the inferred user code to obtain a user ID.